TRADEPASS ACADEMY

309A COMPANION SERIES · VOLUME 5

Flashcard Deck

250 printable flashcards covering all 33 chapters of the Construction Electrician 309A curriculum, with an answer, explanation and memory tip on every card.

- 250 original cards, none repeated from the study guide
- Grouped by chapter, matching your book exactly
- Every card carries a memory tip, not just an answer
- Six embedded reference diagrams for core visual concepts
- Print, cut and carry, or study straight from the page

Companion resource to: Red Seal Construction Electrician 309A Exam Prep, 2026–2027 Edition

Contents

How to use this deck

Each card below is printed front, then back, one after another rather than on a physical two-sided card, so the deck can be studied straight from these pages or cut and pasted onto index cards for portable review. Cover the "Back" callout with your hand or a sheet of paper, answer from memory, then check yourself.

Cards are grouped by chapter to match your Red Seal Construction Electrician 309A study guide exactly, so you can review alongside whichever chapter you are currently studying, or work the whole deck in sequence during final review.

STUDY TIP

In your final week, shuffle your review order rather than always going chapter by chapter. Sequence memory ("I know this one because it follows that one") can create false confidence; true recall is tested when a card appears out of its usual order.

Chapter 1: Introduction to the Trade, the Standard and Exam Strategy

5 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 1.1

FRONT — QUESTION

What document is the 309A exam actually built from?

BACK — ANSWER & MEMORY TIP

Answer: The Red Seal Occupational Standard for Construction Electrician, which lists every task and sub-task the trade covers.

Memory tip: The Standard is the exam's blueprint, not the textbook.

Card 1.2

FRONT — QUESTION

What is the Interprovincial Standards Red Seal program designed to do?

BACK — ANSWER & MEMORY TIP

Answer: Let a certified tradesperson work anywhere in Canada under one recognized credential, without re-certifying province to province.

Memory tip: Red Seal = interprovincial mobility.

Card 1.3

FRONT — QUESTION

On a multiple-choice trade exam, what should you do with a question you cannot answer confidently?

BACK — ANSWER & MEMORY TIP

Answer: Eliminate clearly wrong options, make your best educated guess, flag it, and move on rather than losing time.

Memory tip: An unanswered question is a guaranteed zero.

Card 1.4

FRONT — QUESTION

Why do exam questions often describe a job-site scenario instead of asking a formula directly?

BACK — ANSWER & MEMORY TIP

Answer: Because the exam tests applied judgment, not just recall; you must translate the scenario into the right rule or calculation.

Memory tip: Read for what's actually being asked, not just the numbers.

Card 1.5

FRONT — QUESTION

What is the value of a diagnostic practice test taken before a structured study plan begins?

BACK — ANSWER & MEMORY TIP

Answer: It reveals which topic areas are already strong and which need the most study time, so a plan can be built around real gaps.

Memory tip: Study the gap, not the whole book equally.

Chapter 2: Workplace Safety and the Control of Hazardous Energy

10 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 2.1

FRONT — QUESTION

What are the three rights every worker has under occupational health and safety law?

BACK — ANSWER & MEMORY TIP

Answer: The right to know about workplace hazards, the right to participate in identifying and resolving them, and the right to refuse unsafe work.

Memory tip: Know, participate, refuse.

Card 2.2

FRONT — QUESTION

What is the hierarchy of controls, from most to least effective?

BACK — ANSWER & MEMORY TIP

Answer: Elimination, substitution, engineering controls, administrative controls, and personal protective equipment (PPE) last.

Memory tip: PPE is the last line of defence, not the first.

Card 2.3

FRONT — QUESTION

What does WHMIS 2015 harmonize Canadian workplace chemical labelling with?

BACK — ANSWER & MEMORY TIP

Answer: The Globally Harmonized System (GHS), aligning Canadian pictograms, labels and safety data sheets with international standards.

Memory tip: 2015 = GHS alignment.

Card 2.4

FRONT — QUESTION

What is the correct order of the lockout process before working on equipment?

BACK — ANSWER & MEMORY TIP

Answer: Notify affected workers, shut down, isolate all energy sources, lock and tag, then verify zero energy by testing.

Memory tip: Isolate, lock, tag, THEN test.

Card 2.5

FRONT — QUESTION

Why is 'test before touch' a required final step even after lockout is applied?

BACK — ANSWER & MEMORY TIP

Answer: Because a lock only prevents re-energization; it does not prove the circuit is actually de-energized, so a live test on a known source, then the circuit, confirms zero energy.

Memory tip: A locked source can still fail; testing proves it.

Card 2.6

FRONT — QUESTION

What governs the required approach distances near energized equipment in Canada?

BACK — ANSWER & MEMORY TIP

Answer: CSA Z462, the workplace electrical safety standard, sets arc-flash and shock approach boundaries.

Memory tip: CSA Z462 = the 'how you work near it' standard.

Card 2.7

FRONT — QUESTION

What is a hazardous location, in code terms?

BACK — ANSWER & MEMORY TIP

Answer: An area where flammable gases, vapours, dust or fibres may be present in quantities capable of causing fire or explosion, requiring specially rated equipment.

Memory tip: Classified by what's in the air, not just what's stored there.

Card 2.8

FRONT — QUESTION

What is the correct sequence for proving a circuit dead with a meter?

BACK — ANSWER & MEMORY TIP

Answer: Test the meter on a known live source, test the target circuit, then re-test the meter on the known live source again.

Memory tip: Prove the meter twice; test the circuit once, in between.

Card 2.9

FRONT — QUESTION

What is a designated substance under Canadian OHS law?

BACK — ANSWER & MEMORY TIP

Answer: A specific hazardous material (such as asbestos or lead) that carries its own regulated exposure limits and control requirements beyond general WHMIS rules.

Memory tip: Designated substances get their own dedicated regulation, not just a WHMIS label.

Card 2.10

FRONT — QUESTION

Why does PPE selection for arc flash require an incident-energy assessment rather than a guess?

BACK — ANSWER & MEMORY TIP

Answer: Because arc-flash energy varies by equipment, fault current and clearing time, so the correct PPE category must be calculated or read from labelled equipment data, not assumed.

Memory tip: PPE category comes from data, not habit.

[FIGURE FC-2.1 — ILLUSTRATION PLACEHOLDER]

Caption: A worker in full PPE performing a lockout/tagout procedure on an electrical panel.

Placement: Beside the lockout/tagout card group.

Image generation prompt: *Photorealistic safety-training illustration of a electrician in arc-flash rated PPE (helmet with face shield, insulated gloves, flame-resistant coveralls) applying a lockout padlock and tag to an open electrical panel disconnect. Clean, professional, instructional tone, neutral industrial background, no text overlay, safety-manual quality.*

Chapter 3: Tools, Equipment, Access and Rigging

6 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 3.1

FRONT — QUESTION

Why must insulated hand tools be visually inspected before each use near live parts?

BACK — ANSWER & MEMORY TIP

Answer: Because the insulating coating can develop nicks, cuts or contamination that compromise its rated protection, and only a fresh visual check catches that.

Memory tip: Insulation rating means nothing if the coating is damaged.

Card 3.2

FRONT — QUESTION

What distinguishes a Category II insulated tool rating from Category III?

BACK — ANSWER & MEMORY TIP

Answer: Both relate to working voltage protection levels, with higher categories rated for higher working voltages and greater transient overvoltage protection; always match the tool category to the system voltage.

Memory tip: Match the tool's category to the voltage you're working near.

Card 3.3

FRONT — QUESTION

What is the purpose of a fall-arrest system versus a fall-restraint system?

BACK — ANSWER & MEMORY TIP

Answer: Fall restraint prevents a worker from reaching a fall hazard at all; fall arrest safely stops a fall that has already begun.

Memory tip: Restraint prevents the fall; arrest survives it.

Card 3.4

FRONT — QUESTION

Why is a rigging load always compared against its working load limit, not its breaking strength?

BACK — ANSWER & MEMORY TIP

Answer: The working load limit already includes a safety factor below the point of failure; loading to breaking strength leaves no margin at all.

Memory tip: Rated capacity already has the safety margin built in.

Card 3.5

FRONT — QUESTION

What must be confirmed before using a portable power tool in a damp or wet location?

BACK — ANSWER & MEMORY TIP

Answer: That it is double-insulated or supplied through GFCI protection appropriate for the environment.

Memory tip: Wet location = GFCI or double-insulated, no exceptions.

Card 3.6

FRONT — QUESTION

Why are hoisting and rigging duties often restricted to specifically trained and authorized personnel?

BACK — ANSWER & MEMORY TIP

Answer: Because incorrect load calculations or rigging setups can cause equipment failure and serious injury, so competency must be demonstrated, not assumed.

Memory tip: Rigging authority is earned through training, not tenure.

Chapter 4: Reading Drawings, Specifications and Organizing Work

6 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 4.1

FRONT — QUESTION

What is the difference between an electrical drawing's floor plan and its single-line diagram?

BACK — ANSWER & MEMORY TIP

Answer: The floor plan shows physical device locations and routing; the single-line diagram shows the electrical system's logical connections and protection devices in schematic form.

Memory tip: Floor plan = where; single-line = how it's connected.

Card 4.2

FRONT — QUESTION

What does a drawing scale of 1:50 mean in practical terms?

BACK — ANSWER & MEMORY TIP

Answer: One unit on the drawing equals 50 of the same unit in real life, so a 20 mm measurement on paper represents 1 metre on site.

Memory tip: Scale ratio applies to both dimensions, not just length.

Card 4.3

FRONT — QUESTION

What is the purpose of the MasterFormat numbering system in specifications?

BACK — ANSWER & MEMORY TIP

Answer: It organizes construction specifications into standardized divisions (such as Division 26 for electrical) so trades can find their relevant sections consistently across any project.

Memory tip: MasterFormat gives every trade a predictable place to look.

Card 4.4

FRONT — QUESTION

Why must an electrician cross-reference the drawing legend before interpreting symbols on a new project's plans?

BACK — ANSWER & MEMORY TIP

Answer: Because some symbols vary slightly between designers or firms, and the legend on that specific drawing set is the authoritative key for that project.

Memory tip: Never assume a symbol; the legend is the source of truth for that job.

Card 4.5

FRONT — QUESTION

What information does a drawing's revision block provide?

BACK — ANSWER & MEMORY TIP

Answer: A history of changes made to the drawing, including the date and description of each revision, so the current version can be confirmed as the latest.

Memory tip: Always work from the newest revision, confirmed against the block.

Card 4.6

FRONT — QUESTION

Why is organizing work into a sequence of tasks important on a multi-trade site?

BACK — ANSWER & MEMORY TIP

Answer: Because electrical work often depends on the completion state of other trades (framing, drywall, mechanical), and poor sequencing causes rework and delays.

Memory tip: Sequencing avoids installing conduit that gets buried by the next trade's work.

Chapter 5: Support Structures and Fastening Systems

5 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 5.1

FRONT — QUESTION

Why does the correct fastener choice depend on the base material, not just the load?

BACK — ANSWER & MEMORY TIP

Answer: Because a fastener rated for solid concrete will fail differently in hollow block, wood, or steel; the base material governs the fastener's actual holding capacity.

Memory tip: Same fastener, different base material, different real-world capacity.

Card 5.2

FRONT — QUESTION

What is a strut-channel support system typically used for?

BACK — ANSWER & MEMORY TIP

Answer: Creating a rigid framework to support conduit, cable tray, panelboards or other electrical equipment, especially where structural attachment points are limited.

Memory tip: Strut channel builds you a mounting point where none existed.

Card 5.3

FRONT — QUESTION

What must be considered when sizing a support for a suspended electrical load?

BACK — ANSWER & MEMORY TIP

Answer: The total weight of the load plus its supported components, spacing between supports, and an appropriate safety factor below the fastener's rated capacity.

Memory tip: Size to the total system weight, not just the visible equipment.

Card 5.4

FRONT — QUESTION

Why is seismic restraint required for certain electrical installations in some jurisdictions?

BACK — ANSWER & MEMORY TIP

Answer: To prevent equipment from shifting, falling, or damaging connected systems during an earthquake, particularly for critical or heavy suspended equipment.

Memory tip: Seismic bracing protects both the equipment and anything below it.

Card 5.5

FRONT — QUESTION

What is the general decision process for choosing a fastener on a job?

BACK — ANSWER & MEMORY TIP

Answer: Identify the base material, determine the load and direction of force, select a fastener rated for both, and confirm the required safety factor is met.

Memory tip: Base material and load direction both drive the choice.

Chapter 6: Commissioning and Decommissioning Electrical Systems

5 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 6.1

FRONT — QUESTION

What does an insulation-resistance test measure, and what instrument performs it?

BACK — ANSWER & MEMORY TIP

Answer: It measures the resistance of insulation between conductors or to ground, using a megohmmeter (megger), to confirm insulation is sound before energizing.

Memory tip: Megger tests insulation; it is never used on a live circuit.

Card 6.2

FRONT — QUESTION

What is the purpose of functional testing during commissioning?

BACK — ANSWER & MEMORY TIP

Answer: To confirm that a system operates as designed once energized, including sequencing, protection response and control logic, not just that it is wired correctly.

Memory tip: Wired correctly and functioning correctly are two different confirmations.

Card 6.3

FRONT — QUESTION

Why must decommissioning be documented as thoroughly as commissioning?

BACK — ANSWER & MEMORY TIP

Answer: So future workers know a system or circuit has been safely de-energized and isolated, preventing confusion or unsafe assumptions later.

Memory tip: Documentation protects the next person who touches the system.

Card 6.4

FRONT — QUESTION

What closes both the commissioning and decommissioning process?

BACK — ANSWER & MEMORY TIP

Answer: Complete, accurate documentation confirming the system's final state, test results, and any relevant sign-offs.

Memory tip: No process is finished until it's documented.

Card 6.5

FRONT — QUESTION

Why is an insulation-resistance test performed before initial energization of new wiring?

BACK — ANSWER & MEMORY TIP

Answer: To catch damaged insulation, moisture ingress, or wiring faults before power is applied, preventing an in-service failure or hazard.

Memory tip: Better to catch a fault with a megger than with a breaker trip.

Chapter 7: Communication, Mentoring and Essential Skills

3 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 7.1

FRONT — QUESTION

What are 'essential skills' in the context of a skilled trade?

BACK — ANSWER & MEMORY TIP

Answer: The foundational abilities (reading, numeracy, communication, problem-solving) that underlie safe and competent performance of trade-specific tasks.

Memory tip: Essential skills are the base every technical skill is built on.

Card 7.2

FRONT — QUESTION

Why is clear communication on a jobsite considered a safety issue, not just a courtesy?

BACK — ANSWER & MEMORY TIP

Answer: Because misunderstood instructions near energized equipment or heavy loads can directly cause injury; clarity prevents dangerous assumptions.

Memory tip: On a jobsite, unclear communication is a hazard, not just an inconvenience.

Card 7.3

FRONT — QUESTION

What is the role of a mentor to an apprentice in the trade?

BACK — ANSWER & MEMORY TIP

Answer: To transfer practical knowledge, model safe and professional habits, and help the apprentice build competence and judgment beyond what a classroom teaches.

Memory tip: Mentoring passes down judgment, not just technique.

Chapter 8: Electrical Fundamentals, Ohm's Law, Power and DC Circuits

12 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 8.1

FRONT — QUESTION

What are the four electrical quantities and their units?

BACK — ANSWER & MEMORY TIP

Answer: Voltage (volts), current (amperes), resistance (Ω), and power (watts).

Memory tip: Four quantities, four units, one relationship (Ohm's law) tying three of them together.

Card 8.2

FRONT — QUESTION

State Ohm's law in its basic form.

BACK — ANSWER & MEMORY TIP

Answer: E = I x R: voltage equals current multiplied by resistance.

Memory tip: Cover the unknown in the E-I-R triangle to find any formula variant.

Card 8.3

FRONT — QUESTION

How is electrical power calculated from voltage and current?

BACK — ANSWER & MEMORY TIP

Answer: P = E x I: power in watts equals voltage in volts multiplied by current in amperes.

Memory tip: Add resistance and you get P = I squared R or $P = E^2 / R$.

Card 8.4

FRONT — QUESTION

In a series DC circuit, how does current behave through each component?

BACK — ANSWER & MEMORY TIP

Answer: Current is identical at every point in the circuit, since there is only one path for it to flow.

Memory tip: Series: one path, one current value everywhere.

Card 8.5

FRONT — QUESTION

In a series DC circuit, how does voltage behave across multiple components?

BACK — ANSWER & MEMORY TIP

Answer: The source voltage divides across each component proportional to its resistance; the drops sum to the source voltage.

Memory tip: Voltage divides in series; the pieces always add back to the whole.

Card 8.6

FRONT — QUESTION

In a parallel DC circuit, what stays the same across every branch?

BACK — ANSWER & MEMORY TIP

Answer: The voltage across each parallel branch is identical and equal to the source voltage.

Memory tip: Parallel: same voltage, different currents per branch.

Card 8.7

FRONT — QUESTION

How does total resistance change as more resistors are added in parallel?

BACK — ANSWER & MEMORY TIP

Answer: Total resistance always decreases, and is always smaller than the smallest individual branch resistance.

Memory tip: More parallel paths, less overall opposition to current.

Card 8.8

FRONT — QUESTION

What is the correct way to connect a voltmeter to measure voltage across a component?

BACK — ANSWER & MEMORY TIP

Answer: In parallel with the component, since a voltmeter has very high internal resistance and would block current if placed in series.

Memory tip: Voltmeter: parallel. Ammeter: series. Mixing them up damages the meter.

Card 8.9

FRONT — QUESTION

What is the correct way to connect an ammeter to measure circuit current?

BACK — ANSWER & MEMORY TIP

Answer: In series with the circuit, since an ammeter has very low internal resistance and must carry the full circuit current to read it.

Memory tip: An ammeter placed in parallel by mistake creates a near short circuit.

Card 8.10

FRONT — QUESTION

What happens to voltage readings across a working component when another component in the same series circuit opens (fails)?

BACK — ANSWER & MEMORY TIP

Answer: The full source voltage appears across the open component, and the voltage across all remaining good components drops to zero, since no current flows.

Memory tip: No current, no voltage drop across the good parts; all the voltage stacks on the open point.

Card 8.11

FRONT — QUESTION

Why does a loose or corroded connection in a circuit generate heat?

BACK — ANSWER & MEMORY TIP

Answer: The added resistance at the poor connection causes I^2R heating concentrated at that single point, even though total circuit current may be normal.

Memory tip: A tiny added resistance in the wrong place still burns real power as heat.

Card 8.12

FRONT — QUESTION

What is the formula for equal resistors in parallel?

BACK — ANSWER & MEMORY TIP

Answer: Total resistance equals one resistor's value divided by the number of resistors: $RT = R / n$.

Memory tip: Three equal 30-ohm resistors in parallel give 10 Ω total.

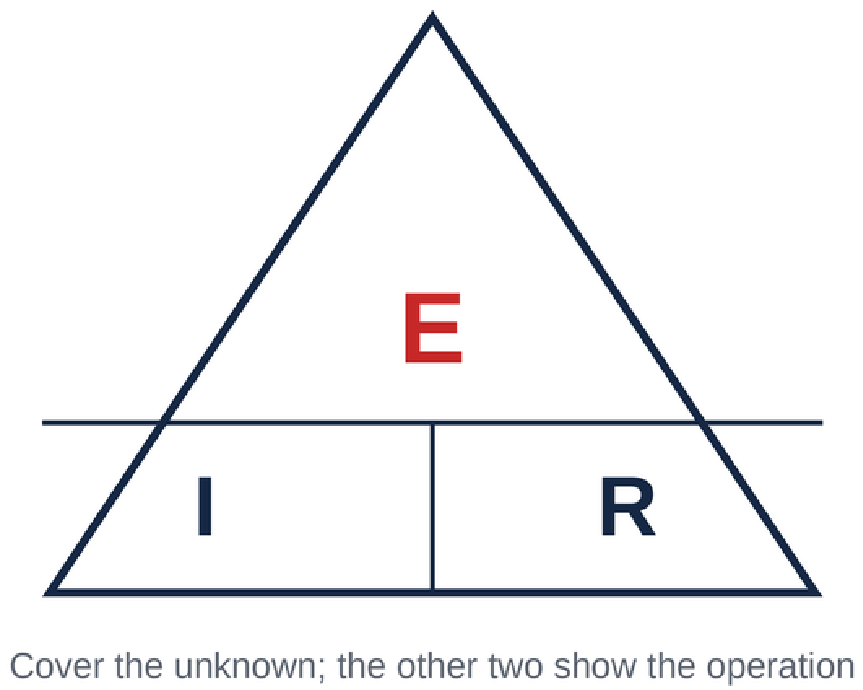

Figure FC-8.1 — The Ohm's law triangle: cover the unknown quantity to find its formula.

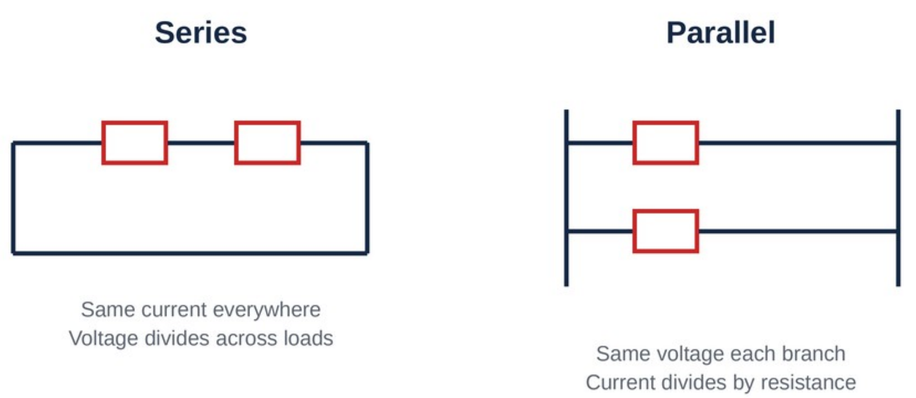

Figure FC-8.2 — Series versus parallel circuit behaviour compared.

Chapter 9: AC Theory, Single-Phase, Power Factor and Vector Reasoning

12 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 9.1

FRONT — QUESTION

What determines the frequency of an AC waveform, and what is Canada's standard utility frequency?

BACK — ANSWER & MEMORY TIP

Answer: Frequency is the number of complete cycles per second; Canada's standard is 60 Hz.

Memory tip: 60 cycles every second, every outlet, everywhere in Canada.

Card 9.2

FRONT — QUESTION

What is the relationship between peak voltage and RMS (effective) voltage?

BACK — ANSWER & MEMORY TIP

Answer: Peak voltage equals RMS voltage multiplied by 1.414; RMS equals peak multiplied by 0.707.

Memory tip: Meters read RMS; peak is always the larger number.

Card 9.3

FRONT — QUESTION

Why do meters display RMS values instead of peak values?

BACK — ANSWER & MEMORY TIP

Answer: Because RMS represents the equivalent heating or working value compared to DC, which is what matters for rating and sizing equipment.

Memory tip: RMS is the 'effective' value: what the load actually experiences as useful power.

Card 9.4

FRONT — QUESTION

What is reactance, and what are its two types?

BACK — ANSWER & MEMORY TIP

Answer: Reactance is AC opposition to current from inductance or capacitance; inductive reactance (XL) and capacitive reactance (XC).

Memory tip: Resistance opposes always; reactance opposes only in AC.

Card 9.5

FRONT — QUESTION

What happens to inductive reactance as frequency increases?

BACK — ANSWER & MEMORY TIP

Answer: Inductive reactance increases proportionally with frequency, since XL = 2 x pi x f x L.

Memory tip: Higher frequency, more opposition from an inductor.

Card 9.6

FRONT — QUESTION

What happens to capacitive reactance as frequency increases?

BACK — ANSWER & MEMORY TIP

Answer: Capacitive reactance decreases as frequency increases, since XC = 1 / (2 x pi x f x C).

Memory tip: Capacitors pass high frequencies more easily; inductors resist them more.

Card 9.7

FRONT — QUESTION

In a purely inductive AC circuit, how does current relate to voltage in phase?

BACK — ANSWER & MEMORY TIP

Answer: Current lags voltage by 90 degrees.

Memory tip: ELI the ICE man: in an inductor (L), voltage (E) leads current (I).

Card 9.8

FRONT — QUESTION

In a purely capacitive AC circuit, how does current relate to voltage in phase?

BACK — ANSWER & MEMORY TIP

Answer: Current leads voltage by 90 degrees.

Memory tip: ELI the ICE man: in a capacitor (C), current (I) leads voltage (E).

Card 9.9

FRONT — QUESTION

What is power factor, expressed as a formula?

BACK — ANSWER & MEMORY TIP

Answer: Power factor equals true power divided by apparent power: PF = P (watts) / S (volt-amperes), also equal to the cosine of the phase angle.

Memory tip: PF can never be greater than 1.

Card 9.10

FRONT — QUESTION

Why does a low power factor matter to a utility and to equipment sizing, even though it doesn't change the true power delivered?

BACK — ANSWER & MEMORY TIP

Answer: A low power factor means higher current is needed to deliver the same true power, increasing conductor and equipment sizing and utility system losses.

Memory tip: Low PF: more current for the same useful work, which costs everyone more.

Card 9.11

FRONT — QUESTION

What corrects a lagging (inductive) power factor in an electrical system?

BACK — ANSWER & MEMORY TIP

Answer: Adding capacitance, typically via power-factor-correction capacitor banks, which offsets the inductive reactance.

Memory tip: Capacitors correct a lagging PF caused by motors and other inductive loads.

Card 9.12

FRONT — QUESTION

What is impedance, and how is it calculated in a simple series RL or RC circuit?

BACK — ANSWER & MEMORY TIP

Answer: Impedance (Z) is total AC opposition combining resistance and reactance: Z = square root of (R^2 + X^2).

Memory tip: Impedance is Ohm's law's AC cousin, folding in reactance.

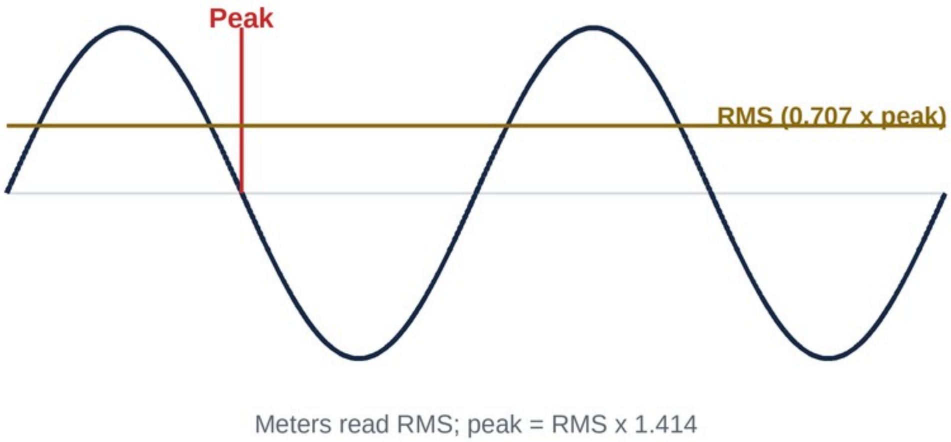

Figure FC-9.1 — An AC sine wave showing peak and RMS (effective) values.

Chapter 10: Three-Phase Systems, Wye, Delta and Line-versus-Phase

12 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 10.1

FRONT — QUESTION

What is the fundamental advantage of three-phase power over single-phase for industrial and commercial loads?

BACK — ANSWER & MEMORY TIP

Answer: It delivers more power for a given conductor size, provides smoother, more constant power delivery, and is required for standard three-phase motors.

Memory tip: Three-phase: more efficient delivery, smoother torque for motors.

Card 10.2

FRONT — QUESTION

In a wye (star) connection, how is line voltage related to phase voltage?

BACK — ANSWER & MEMORY TIP

Answer: Line voltage equals phase voltage multiplied by the √3 (approximately 1.732).

Memory tip: Wye: voltage steps up by √3 from phase to line.

Card 10.3

FRONT — QUESTION

In a wye connection, how does line current compare to phase current?

BACK — ANSWER & MEMORY TIP

Answer: They are equal; line current and phase current are the same value in a wye connection.

Memory tip: Wye: current stays the same, voltage changes.

Card 10.4

FRONT — QUESTION

In a delta connection, how is line current related to phase current?

BACK — ANSWER & MEMORY TIP

Answer: Line current equals phase current multiplied by the √3.

Memory tip: Delta is the mirror of wye: current steps up by √3 instead of voltage.

Card 10.5

FRONT — QUESTION

In a delta connection, how does line voltage compare to phase voltage?

BACK — ANSWER & MEMORY TIP

Answer: They are equal; line voltage and phase voltage are the same value in a delta connection.

Memory tip: Delta: voltage stays the same, current changes.

Card 10.6

FRONT — QUESTION

Why does a wye connection provide a neutral point but a delta connection does not?

BACK — ANSWER & MEMORY TIP

Answer: A wye connection joins all three windings at a common centre point, which can be grounded and brought out as a neutral; a delta forms a closed loop with no such centre point.

Memory tip: The letter Y has a centre; the letter shape of delta (a triangle) doesn't.

Card 10.7

FRONT — QUESTION

How do you reverse the rotation direction of a three-phase motor?

BACK — ANSWER & MEMORY TIP

Answer: Interchange any two of the three line conductors feeding it.

Memory tip: Swap two, reverse the rotation, every time.

Card 10.8

FRONT — QUESTION

What is the three-phase power formula using line values and power factor?

BACK — ANSWER & MEMORY TIP

Answer: $P = \sqrt{3} \times E_L \times I_L \times PF$ (true power in watts, using line voltage and line current).

Memory tip: Every three-phase power formula carries the $\sqrt{3}$ factor.

Card 10.9

FRONT — QUESTION

Why is 120/208 V common in Canadian commercial buildings while 347/600 V is common in larger industrial or commercial services?

BACK — ANSWER & MEMORY TIP

Answer: 120/208 V wye serves standard receptacle and lighting loads directly from a low-voltage system; 347/600 V is used for larger loads and long feeders where higher voltage reduces current and conductor size, with local step-down for lower voltage needs.

Memory tip: Bigger buildings, higher distribution voltage, less current to carry.

Card 10.10

FRONT — QUESTION

What is a balanced three-phase load?

BACK — ANSWER & MEMORY TIP

Answer: A load where all three phases draw equal current, keeping the system evenly loaded and the neutral current (in a wye system) near zero.

Memory tip: Balanced = equal draw on all three legs.

Card 10.11

FRONT — QUESTION

Why can an unbalanced three-phase load cause neutral current to rise significantly?

BACK — ANSWER & MEMORY TIP

Answer: Because the neutral carries the vector difference of unequal phase currents; the more unequal the phases, the more current returns on the neutral.

Memory tip: Unbalance shows up as current on a conductor that should carry very little.

Card 10.12

FRONT — QUESTION

What is the practical significance of the √3 (1.732) factor appearing in every three-phase calculation?

BACK — ANSWER & MEMORY TIP

Answer: It accounts for the 120-degree phase displacement between the three voltages, and omitting it is one of the most common calculation errors on three-phase problems.

Memory tip: Forgetting √3 is the single most common three-phase math mistake.

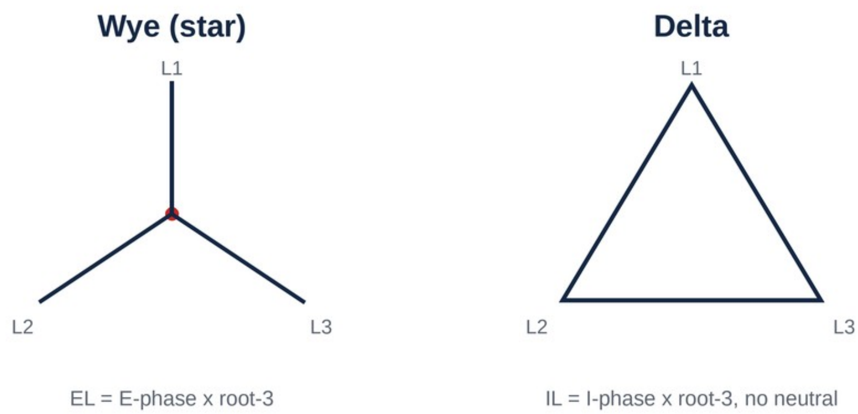

Figure FC-10.1 — Wye and delta connections compared, line versus phase quantities.

Chapter 11: Consumer and Supply Services and Metering

7 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 11.1

FRONT — QUESTION

What are the three main components found at a typical service entrance?

BACK — ANSWER & MEMORY TIP

Answer: The service conductors, the service disconnecting means, and the metering equipment.

Memory tip: Conductors bring it in, the meter measures it, the disconnect controls it.

Card 11.2

FRONT — QUESTION

What is the standard configuration of a single-phase, three-wire residential service?

BACK — ANSWER & MEMORY TIP

Answer: Two ungrounded (hot) conductors and one grounded (neutral) conductor, providing both 120 V (line to neutral) and 240 V (line to line).

Memory tip: Three wires, two voltages available from the same service.

Card 11.3

FRONT — QUESTION

Why must grounding and bonding be established at the service, specifically, rather than anywhere downstream?

BACK — ANSWER & MEMORY TIP

Answer: Because the service is the single point where the system neutral is intentionally connected to ground, establishing the reference point for the entire electrical system.

Memory tip: The service is the one and only place neutral and ground/bond meet.

Card 11.4

FRONT — QUESTION

What is the role of a utility meter in the service?

BACK — ANSWER & MEMORY TIP

Answer: It measures the energy (kWh) consumed by the customer for utility billing purposes, typically installed between the service conductors and the main disconnect.

Memory tip: The meter measures usage; it does not protect the circuit.

Card 11.5

FRONT — QUESTION

How does a three-phase service typically differ in configuration from a single-phase service?

BACK — ANSWER & MEMORY TIP

Answer: It uses three ungrounded conductors (plus often a neutral) to deliver three-phase power, commonly in wye configurations like 120/208 V or 347/600 V for larger loads.

Memory tip: Three-phase: three hot legs, often with a neutral for lower-voltage loads too.

Card 11.6

FRONT — QUESTION

What must be verified about service conductor sizing before a service is energized?

BACK — ANSWER & MEMORY TIP

Answer: That the conductors are rated to carry the calculated service load without exceeding their ampacity, factoring in any required demand factors.

Memory tip: Service conductors must match the calculated load, not just 'look big enough.'

Card 11.7

FRONT — QUESTION

Why is metering equipment typically installed ahead of (upstream of) the main service disconnect?

BACK — ANSWER & MEMORY TIP

Answer: So that all energy passing to the building, including through the disconnect itself, is measured and billed accurately.

Memory tip: The meter must see everything, so it comes first in the sequence.

Chapter 12: Service and Feeder Load Calculations

9 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 12.1

FRONT — QUESTION

What are the three general kinds of load considered in a service or feeder calculation?

BACK — ANSWER & MEMORY TIP

Answer: Continuous loads, non-continuous (general) loads, and loads with special demand factor rules (such as ranges, dryers or heating).

Memory tip: Not all loads are treated the same in a calculation; the type changes the math.

Card 12.2

FRONT — QUESTION

What is the general purpose of the single-dwelling load calculation method?

BACK — ANSWER & MEMORY TIP

Answer: To calculate a realistic minimum service size for a house using standardized base allowances and demand factors, rather than adding up connected load at 100%.

Memory tip: The method reflects real usage patterns, not worst-case simultaneous use.

Card 12.3

FRONT — QUESTION

What does the 'larger of' rule generally compare in a load calculation involving heating and air conditioning?

BACK — ANSWER & MEMORY TIP

Answer: Since heating and cooling are rarely operating at the same time, the calculation uses only the larger of the two loads rather than adding both together.

Memory tip: Heat in winter, cool in summer, never both at once, so only the bigger one counts.

Card 12.4

FRONT — QUESTION

What is the continuous load rule (the eighty-per-cent rule) in a load calculation?

BACK — ANSWER & MEMORY TIP

Answer: A continuous load (operating three hours or more) must not exceed 80% of the overcurrent device or conductor rating, unless the equipment is rated for 100% continuous duty.

Memory tip: Continuous loads get derated to 80% of the device's rating.

Card 12.5

FRONT — QUESTION

Why does a load calculation use demand factors instead of simply summing all connected loads at face value?

BACK — ANSWER & MEMORY TIP

Answer: Because not every load operates simultaneously at full capacity in real use, so demand factors produce a realistic, code-recognized service size instead of an oversized, wasteful one.

Memory tip: Demand factors reflect how buildings are actually used, not worst-case theory.

Card 12.6

FRONT — QUESTION

What is a feeder, as distinct from a branch circuit?

BACK — ANSWER & MEMORY TIP

Answer: A feeder is the conductor set that carries power from the main service or a distribution point to a panelboard or piece of distribution equipment, rather than directly to the final load.

Memory tip: Feeder feeds a panel; branch circuit feeds the actual load.

Card 12.7

FRONT — QUESTION

Why does a worked single-dwelling calculation typically begin with a base allowance before adding specific appliance loads?

BACK — ANSWER & MEMORY TIP

Answer: Because the code recognizes a minimum general lighting and receptacle load for any dwelling, which forms the starting point before additional major loads are layered on.

Memory tip: Start with the base allowance, then add the big-ticket appliances.

Card 12.8

FRONT — QUESTION

What is the effect of applying a demand factor to a group of loads such as multiple dwelling units?

BACK — ANSWER & MEMORY TIP

Answer: It reduces the calculated total below the sum of each unit's individual connected load, recognizing that not all units will be at peak demand simultaneously.

Memory tip: More units, generally a lower percentage demand factor applied to the group.

Card 12.9

FRONT — QUESTION

Why is it critical to correctly classify a load as continuous versus non-continuous in a calculation?

BACK — ANSWER & MEMORY TIP

Answer: Because continuous loads require the 80% derating on protection and conductor sizing, while non-continuous loads generally do not, changing the required conductor and breaker size.

Memory tip: Get the classification wrong, and the sizing math is wrong too.

Chapter 13: Overcurrent, Ground-Fault, Arc-Fault and Surge Protection

9 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 13.1

FRONT — QUESTION

What are the three kinds of overcurrent recognized in protection theory?

BACK — ANSWER & MEMORY TIP

Answer: Overload, short circuit, and ground fault.

Memory tip: Three distinct overcurrent conditions, each with its own protective focus.

Card 13.2

FRONT — QUESTION

What distinguishes a fuse from a circuit breaker in basic operating principle?

BACK — ANSWER & MEMORY TIP

Answer: A fuse is a single-use device that melts an element to open the circuit; a circuit breaker mechanically trips and can be reset and reused.

Memory tip: Fuse: sacrifice itself once. Breaker: trip, reset, reuse.

Card 13.3

FRONT — QUESTION

What is the interrupting rating of an overcurrent device, and why does it matter?

BACK — ANSWER & MEMORY TIP

Answer: It is the maximum fault current the device can safely interrupt without failure; the device's interrupting rating must equal or exceed the available fault current at its installation point.

Memory tip: An underrated device can literally explode under a fault it can't handle.

Card 13.4

FRONT — QUESTION

How does a Class A ground-fault circuit interrupter protect people, specifically?

BACK — ANSWER & MEMORY TIP

Answer: It continuously compares current on the live conductor to current returning on the neutral, and opens the circuit when it detects an imbalance around 4 to 6 milliamperes, which can indicate current leaking through a person.

Memory tip: GFCI watches for a current imbalance, not just overcurrent.

Card 13.5

FRONT — QUESTION

How does arc-fault protection differ in purpose from ground-fault protection?

BACK — ANSWER & MEMORY TIP

Answer: Arc-fault protection detects the electrical signature of dangerous arcing (a fire hazard) rather than a current imbalance to ground (a shock hazard), addressing a different risk.

Memory tip: GFCI protects against shock; AFCI protects against fire from arcing.

Card 13.6

FRONT — QUESTION

Where in a distribution system is a Type 1 surge protective device typically installed, compared to a Type 2 device?

BACK — ANSWER & MEMORY TIP

Answer: A Type 1 SPD is installed ahead of the main service disconnect (utility side or main), while a Type 2 SPD is installed downstream at distribution panels; both limit transient overvoltage but at different points in the system.

Memory tip: Type 1 guards the front door; Type 2 guards the rooms further inside.

Card 13.7

FRONT — QUESTION

Why can two very different devices both be described using the term 'ground-fault protection'?

BACK — ANSWER & MEMORY TIP

Answer: Because ground-fault protection exists at different scales: a Class A GFCI protects people at very low current levels, while ground-fault protection on larger equipment protects equipment and systems at much higher current thresholds.

Memory tip: Same term, different purpose and threshold depending on scale.

Card 13.8

FRONT — QUESTION

What is the basic operating difference between how a fuse and a thermal-magnetic breaker respond to a short circuit versus an overload?

BACK — ANSWER & MEMORY TIP

Answer: Both typically have a fast magnetic (or equivalent) response to high short-circuit currents and a slower thermal response to sustained overload currents, giving a time-current curve with two distinct regions.

Memory tip: Fast response to a big fault; slower response to a smaller sustained overload.

Card 13.9

FRONT — QUESTION

What must be confirmed about an overcurrent device's rating relative to the conductor it protects?

BACK — ANSWER & MEMORY TIP

Answer: That the device's ampere rating does not exceed the conductor's ampacity, except where specific code exceptions apply (such as motor circuits).

Memory tip: Generally, protect the conductor at or below its rated ampacity.

Chapter 14: Power Distribution Equipment

8 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 14.1

FRONT — QUESTION

What is the function of a panelboard in an electrical distribution system?

BACK — ANSWER & MEMORY TIP

Answer: It receives feeder or service power and distributes it to multiple branch circuits, each protected by its own overcurrent device.

Memory tip: Panelboard: one feed in, many protected circuits out.

Card 14.2

FRONT — QUESTION

How does a switchboard generally differ from a panelboard?

BACK — ANSWER & MEMORY TIP

Answer: A switchboard is typically larger, front- or rear-accessible, and handles higher current distribution for larger commercial or industrial systems, while a panelboard is smaller and enclosure-mounted.

Memory tip: Switchboard: bigger scale distribution. Panelboard: branch-circuit level.

Card 14.3

FRONT — QUESTION

What is busway, and where is it typically used?

BACK — ANSWER & MEMORY TIP

Answer: A busway is a prefabricated metal-enclosed assembly of busbars used to distribute large amounts of power efficiently, often in industrial or high-rise applications.

Memory tip: Busway moves big power efficiently along a run, instead of massive conduit and cable.

Card 14.4

FRONT — QUESTION

What is the function of a disconnecting means for a piece of equipment?

BACK — ANSWER & MEMORY TIP

Answer: To provide a means of fully isolating the equipment from its power source for maintenance, service, or emergency shutdown.

Memory tip: A disconnect must be able to fully de-energize what it serves.

Card 14.5

FRONT — QUESTION

What is the purpose of a transfer switch in a power system with a standby generator?

BACK — ANSWER & MEMORY TIP

Answer: To safely switch a load between the normal utility source and the standby source, preventing both from being connected simultaneously (which could backfeed the utility).

Memory tip: A transfer switch prevents two sources from ever meeting at once.

Card 14.6

FRONT — QUESTION

Why is dedicated working space required in front of electrical equipment such as panelboards?

BACK — ANSWER & MEMORY TIP

Answer: To ensure workers have safe, unobstructed room to operate, inspect, or service the equipment without being forced into an unsafe position near energized parts.

Memory tip: Working space isn't storage space; it must stay clear.

Card 14.7

FRONT — QUESTION

What must be verified about a transfer switch's rating relative to the loads it serves?

BACK — ANSWER & MEMORY TIP

Answer: That its ampere rating and configuration (automatic or manual) are appropriate for the connected load and the switching sequence required by the system design.

Memory tip: A transfer switch must be sized and configured to match the actual load and system.

Card 14.8

FRONT — QUESTION

Why might a switchboard include multiple sections rather than a single enclosure?

BACK — ANSWER & MEMORY TIP

Answer: To accommodate distribution requirements such as metering, main protection, and multiple feeder or branch sections within a large, organized assembly.

Memory tip: Bigger systems, more organized sections within the switchboard.

Chapter 15: Power Quality and Standby Systems

6 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 15.1

FRONT — QUESTION

What are power-quality disturbances, broadly speaking?

BACK — ANSWER & MEMORY TIP

Answer: Deviations from ideal, clean sinusoidal power, including sags, swells, transients, harmonics, and interruptions.

Memory tip: Power quality issues are deviations from the 'perfect' sine wave.

Card 15.2

FRONT — QUESTION

What causes harmonic distortion, and what equipment commonly generates it?

BACK — ANSWER & MEMORY TIP

Answer: Non-linear loads (such as variable-frequency drives, electronic ballasts, and switch-mode power supplies) draw current in a non-sinusoidal pattern, generating harmonic currents.

Memory tip: Electronics draw power in 'chunks,' not a smooth wave, creating harmonics.

Card 15.3

FRONT — QUESTION

Why can harmonics cause elevated current on the neutral conductor of a three-phase system, even under a balanced fundamental load?

BACK — ANSWER & MEMORY TIP

Answer: Certain harmonic orders (particularly the third and its multiples) do not cancel at the neutral the way fundamental balanced currents do, and can add together instead.

Memory tip: Harmonics can defeat the usual neutral cancellation in three-phase systems.

Card 15.4

FRONT — QUESTION

What is the function of an uninterruptible power supply (UPS)?

BACK — ANSWER & MEMORY TIP

Answer: To provide immediate, brief backup power (typically from batteries) during a utility interruption, bridging the gap until a generator starts or power is restored.

Memory tip: UPS: instant bridge power, not long-term standby.

Card 15.5

FRONT — QUESTION

How does an uninterruptible power supply differ in role from a standby generator?

BACK — ANSWER & MEMORY TIP

Answer: A UPS provides near-instant, short-duration backup to prevent any interruption at all; a generator provides longer-duration backup power but takes time to start and transfer.

Memory tip: UPS covers the gap; the generator covers the duration.

Card 15.6

FRONT — QUESTION

What is the general function of a power-conditioning device?

BACK — ANSWER & MEMORY TIP

Answer: To improve power quality by filtering, regulating, or correcting disturbances such as voltage sags, surges, or harmonics before they reach sensitive equipment.

Memory tip: Power conditioning cleans up the power before equipment sees it.

Chapter 16: Bonding, Grounding and Lightning Protection

8 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 16.1

FRONT — QUESTION

What is the fundamental difference between grounding and bonding?

BACK — ANSWER & MEMORY TIP

Answer: Grounding is the intentional connection of the electrical system to earth through a grounding electrode; bonding is the connection of non-current-carrying metal parts together to provide a low-impedance fault current path.

Memory tip: Grounding references the system to earth; bonding clears a fault by carrying current back to the source.

Card 16.2

FRONT — QUESTION

How does bonding actually clear a fault?

BACK — ANSWER & MEMORY TIP

Answer: By providing a low-impedance path back to the source, allowing enough fault current to flow to operate the overcurrent protective device quickly.

Memory tip: Bonding's whole job is to make the breaker trip fast, not to 'go to ground.'

Card 16.3

FRONT — QUESTION

What is a grounding electrode, and give an example.

BACK — ANSWER & MEMORY TIP

Answer: A conductive object (such as a ground rod, building steel, or a concrete-encased electrode) used to establish an intentional connection between the electrical system and the earth.

Memory tip: The electrode is the physical connection point into the earth itself.

Card 16.4

FRONT — QUESTION

What does single-point bonding mean in the context of a service?

BACK — ANSWER & MEMORY TIP

Answer: That the connection between the system neutral and the bonding/grounding system is made at only one point, typically the service, and kept separate everywhere downstream.

Memory tip: One join point only; multiple bonding points downstream create dangerous parallel neutral paths.

Card 16.5

FRONT — QUESTION

Why must equipment bonding maintain continuity along its entire path back to the source?

BACK — ANSWER & MEMORY TIP

Answer: Because a break anywhere in the bonding path prevents fault current from returning effectively, which can leave equipment energized at a dangerous touch voltage during a fault.

Memory tip: A bonding path is only as good as its weakest, most broken link.

Card 16.6

FRONT — QUESTION

What is the purpose of a lightning protection system on a structure?

BACK — ANSWER & MEMORY TIP

Answer: To provide a preferred, low-impedance path for a lightning strike to follow safely to earth, protecting the structure and its occupants from fire and electrical damage.

Memory tip: Lightning protection gives the strike somewhere safe to go.

Card 16.7

FRONT — QUESTION

Why is it incorrect to say a ground-fault current 'goes into the earth' to clear a fault in a typical building system?

BACK — ANSWER & MEMORY TIP

Answer: Because earth itself has relatively high impedance; the fault current actually needs a low-impedance bonded metallic path back to the source to trip the breaker quickly, not a path through soil.

Memory tip: The soil is a poor conductor; the bonding conductor does the real work.

Card 16.8

FRONT — QUESTION

What happens if the neutral and bonding systems are connected at more than one point downstream of the service?

BACK — ANSWER & MEMORY TIP

Answer: It can create unintended parallel paths for neutral current to flow on bonding conductors and metal building parts, which is a shock and fire hazard.

Memory tip: Multiple neutral-bond connections create stray current paths where none should exist.

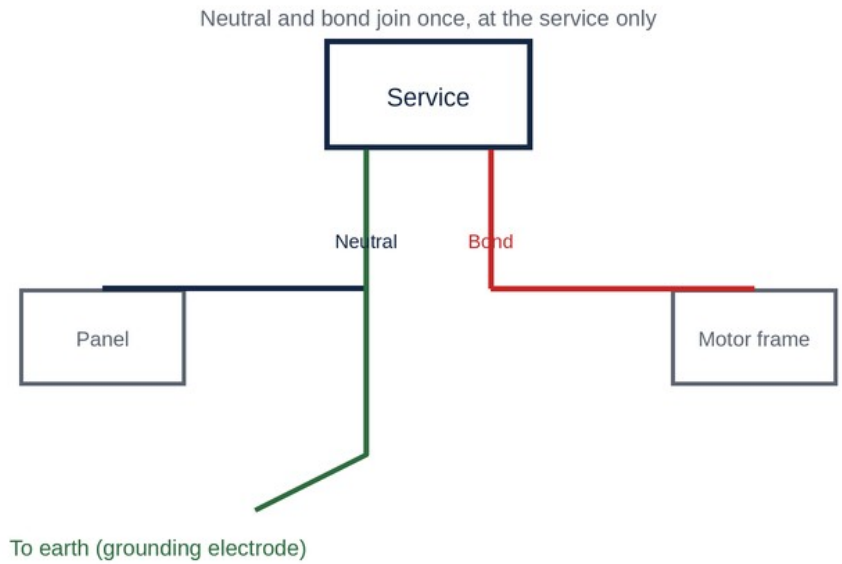

Figure FC-16.1 — Neutral and bond join once, at the service only.

Chapter 17: Power Generation and Conversion

5 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 17.1

FRONT — QUESTION

What is the basic operating principle of an AC generator (alternator)?

BACK — ANSWER & MEMORY TIP

Answer: A rotating magnetic field induces alternating voltage in stationary windings (or vice versa) through electromagnetic induction, producing AC output.

Memory tip: Relative motion between magnetic field and conductor induces voltage.

Card 17.2

FRONT — QUESTION

How does a DC generator differ from an AC generator in basic construction?

BACK — ANSWER & MEMORY TIP

Answer: A DC generator uses a commutator to mechanically switch the output connections, converting the internally generated AC into a pulsed DC output at the terminals.

Memory tip: Same induction principle, but a commutator converts the output to DC.

Card 17.3

FRONT — QUESTION

What is rectification, and what device commonly performs it?

BACK — ANSWER & MEMORY TIP

Answer: Rectification is the conversion of AC to DC, commonly performed using diodes arranged in a rectifier circuit.

Memory tip: Rectifier: AC in, DC out.

Card 17.4

FRONT — QUESTION

What is inversion, and where is it commonly used?

BACK — ANSWER & MEMORY TIP

Answer: Inversion is the conversion of DC to AC, used in applications like solar inverters and uninterruptible power supplies to convert stored or generated DC into usable AC.

Memory tip: Inverter: DC in, AC out, the opposite of a rectifier.

Card 17.5

FRONT — QUESTION

Why might a power conversion system include both rectification and inversion stages?

BACK — ANSWER & MEMORY TIP

Answer: To first convert incoming AC to a stable DC bus, then invert that DC back to a precisely controlled AC output, as used in variable-frequency drives and some UPS designs.

Memory tip: AC to DC to AC gives precise control over the final output.

Chapter 18: Renewable Energy and Energy Storage

6 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 18.1

FRONT — QUESTION

How does a photovoltaic (solar) cell generate electricity?

BACK — ANSWER & MEMORY TIP

Answer: It converts sunlight directly into DC electricity through the photovoltaic effect within semiconductor material.

Memory tip: Solar cells make DC directly from light, no moving parts.

Card 18.2

FRONT — QUESTION

What equipment is required to connect a typical grid-tied solar system to a building's AC system?

BACK — ANSWER & MEMORY TIP

Answer: An inverter, to convert the panels' DC output into grid-compatible AC, along with appropriate disconnects and overcurrent protection.

Memory tip: The inverter is the bridge between solar's native DC and the building's AC.

Card 18.3

FRONT — QUESTION

What hazard is unique to solar photovoltaic systems compared to most other electrical sources?

BACK — ANSWER & MEMORY TIP

Answer: Solar panels are energized any time they are exposed to light, meaning they cannot simply be 'switched off' at the source the way most other equipment can.

Memory tip: You can't de-energize sunlight; the panels stay live whenever lit.

Card 18.4

FRONT — QUESTION

What is the purpose of rapid-shutdown requirements on solar installations?

BACK — ANSWER & MEMORY TIP

Answer: To reduce conductor voltage to a safe level quickly near the array in an emergency, protecting firefighters and other responders.

Memory tip: Rapid shutdown is primarily a first-responder safety measure.

Card 18.5

FRONT — QUESTION

What is the basic function of a battery energy storage system paired with renewable generation?

BACK — ANSWER & MEMORY TIP

Answer: To store excess generated energy for later use, smoothing supply and demand and providing backup power when generation is unavailable.

Memory tip: Storage banks the extra energy for when it's actually needed.

Card 18.6

FRONT — QUESTION

Why must grounding and bonding for renewable energy systems be given specific attention beyond standard wiring practice?

BACK — ANSWER & MEMORY TIP

Answer: Because DC arcing behaves differently than AC arcing, and unique equipment (inverters, combiner boxes, charge controllers) introduces grounding and bonding considerations not present in typical AC-only installations.

Memory tip: DC systems bring their own unique grounding and arcing behaviour.

[FIGURE FC-18.1 — ILLUSTRATION PLACEHOLDER]

Caption: A rooftop solar array with visible rapid-shutdown labelling at the array boundary.

Placement: Beside the solar hazard card group.

Image generation prompt: *Clean technical illustration of a residential rooftop solar photovoltaic array with conduit routing to a combiner box and inverter, a visible rapid-shutdown warning label at the array boundary, daytime, flat illustrative style suitable for a trade textbook, no photorealism required, navy and red accent colours.*

Chapter 19: High-Voltage Systems

4 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 19.1

FRONT — QUESTION

In Canadian electrical code terms, what threshold generally defines 'high voltage'?

BACK — ANSWER & MEMORY TIP

Answer: Voltages exceeding 750 V are generally treated as high voltage under the code, triggering additional requirements.

Memory tip: Above 750 V, a distinct, stricter rule set applies.

Card 19.2

FRONT — QUESTION

Why does high-voltage equipment require specially trained personnel and distinct safety procedures?

BACK — ANSWER & MEMORY TIP

Answer: Because the higher energy levels involved dramatically increase the severity of arc-flash, shock, and equipment failure hazards compared to standard low-voltage work.

Memory tip: Higher voltage doesn't just mean 'more of the same risk,' it changes the risk category entirely.

Card 19.3

FRONT — QUESTION

What is typically required for approach to energized high-voltage equipment, beyond standard low-voltage practice?

BACK — ANSWER & MEMORY TIP

Answer: Larger defined approach boundaries, specialized PPE, and often the use of hot sticks or other insulated tools to maintain safe distance.

Memory tip: High voltage means bigger safety boundaries and more specialized tools.

Card 19.4

FRONT — QUESTION

Why might a facility have its own dedicated high-voltage switching and metering equipment rather than relying solely on utility-side equipment?

BACK — ANSWER & MEMORY TIP

Answer: To manage internal distribution, provide local switching flexibility, and sometimes to reduce utility charges by taking service at a higher voltage and stepping it down internally.

Memory tip: Large facilities often manage their own high-voltage distribution internally.

Chapter 20: Transformers

11 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 20.1

FRONT — QUESTION

What is the basic working principle of a transformer?

BACK — ANSWER & MEMORY TIP

Answer: Mutual electromagnetic induction: alternating current in the primary winding creates a changing magnetic field that induces voltage in the secondary winding.

Memory tip: No physical connection needed between windings, only shared magnetic flux.

Card 20.2

FRONT — QUESTION

What does the turns ratio of a transformer determine?

BACK — ANSWER & MEMORY TIP

Answer: The ratio between primary and secondary voltage: EP / ES = NP / NS.

Memory tip: Voltage follows the turns ratio directly.

Card 20.3

FRONT — QUESTION

How is current related to the turns ratio, compared to voltage?

BACK — ANSWER & MEMORY TIP

Answer: Current follows the inverse of the turns ratio: as voltage steps down, current steps up proportionally, and vice versa.

Memory tip: Voltage and current move in opposite directions across a transformer.

Card 20.4

FRONT — QUESTION

What does 'power is conserved' mean in the context of an ideal transformer?

BACK — ANSWER & MEMORY TIP

Answer: Apparent power in equals apparent power out (EP x IP = ES x IS), aside from real-world losses; a transformer cannot create or destroy power, only change the voltage-current relationship.

Memory tip: A transformer trades voltage for current, but total power stays essentially the same.

Card 20.5

FRONT — QUESTION

Why are transformers rated in kVA rather than kW?

BACK — ANSWER & MEMORY TIP

Answer: Because winding heating depends on current, which relates to kVA regardless of the load's power factor, making kVA the more universal and safety-relevant rating.

Memory tip: kVA rating protects the windings regardless of what power factor the load has.

Card 20.6

FRONT — QUESTION

What is the typical secondary configuration produced by a delta-primary, wye-secondary distribution transformer?

BACK — ANSWER & MEMORY TIP

Answer: A 120/208 V or 347/600 V three-phase wye secondary with a usable neutral, common in commercial distribution.

Memory tip: Delta-wye transformers are how many buildings get their usable neutral.

Card 20.7

FRONT — QUESTION

What is an autotransformer, and how does it differ from a standard two-winding transformer?

BACK — ANSWER & MEMORY TIP

Answer: An autotransformer uses a single winding shared between primary and secondary (with a tap), making it smaller and cheaper, but providing no electrical isolation between input and output.

Memory tip: Autotransformer: shared winding, no isolation, more compact.

Card 20.8

FRONT — QUESTION

What is the purpose of taps on a transformer?

BACK — ANSWER & MEMORY TIP

Answer: To allow small adjustments to the output voltage to compensate for variations in supply voltage or to correct for cable voltage drop.

Memory tip: Taps fine-tune output voltage without needing a different transformer.

Card 20.9

FRONT — QUESTION

What are the two general types of losses in a transformer?

BACK — ANSWER & MEMORY TIP

Answer: Copper losses (I^2R heating in the windings) and iron (core) losses from magnetization of the core material.

Memory tip: Copper losses grow with load; core losses are roughly constant.

Card 20.10

FRONT — QUESTION

Why does transformer polarity matter when paralleling two transformers?

BACK — ANSWER & MEMORY TIP

Answer: Mismatched polarity between paralleled transformers can cause a severe circulating current or a dangerous voltage condition instead of proper load sharing.

Memory tip: Polarity must match before transformers are ever paralleled.

Card 20.11

FRONT — QUESTION

Why must a current transformer's secondary never be opened while its primary is energized?

BACK — ANSWER & MEMORY TIP

Answer: Because with no secondary load path, the transformer attempts to maintain the ampere-turns balance by developing an extremely high, dangerous secondary voltage.

Memory tip: An open CT secondary under load is a serious shock and equipment hazard.

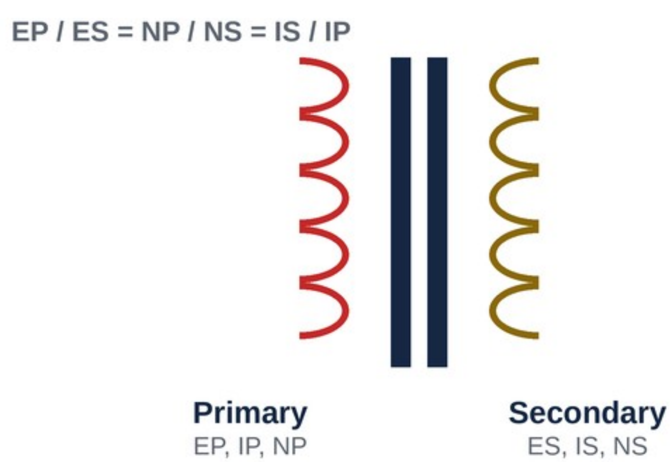

Figure FC-20.1 — Transformer primary and secondary winding relationship.

Chapter 21: Raceways, Conductors, Cables and Enclosures

11 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 21.1

FRONT — QUESTION

What is ampacity?

BACK — ANSWER & MEMORY TIP

Answer: The maximum current a conductor can carry continuously under stated conditions of use without exceeding its temperature rating.

Memory tip: Ampacity is conditional; it changes with temperature, grouping, and insulation.

Card 21.2

FRONT — QUESTION

What is the termination-temperature rule, and why is it heavily tested?

BACK — ANSWER & MEMORY TIP

Answer: Even if a conductor's insulation is rated for a higher temperature (such as 90 °C), the ampacity used for sizing may be limited by the lower temperature rating of its terminations (often 75 °C or 60 °C).

Memory tip: The weakest link in the temperature chain, often the termination, sets the real limit.

Card 21.3

FRONT — QUESTION

What does 'correction and derating' mean when applied to conductor ampacity?

BACK — ANSWER & MEMORY TIP

Answer: Correction factors adjust for ambient temperature different from the table's base temperature; derating factors adjust for the number of current-carrying conductors grouped together.

Memory tip: Correction handles temperature; derating handles conductor count.

Card 21.4

FRONT — QUESTION

What is voltage drop, and why does it matter beyond simple code compliance?

BACK — ANSWER & MEMORY TIP

Answer: It is the loss of voltage along a conductor run due to its resistance; beyond meeting code limits, excessive voltage drop causes poor equipment performance and wasted energy.

Memory tip: Voltage drop isn't just a code number, it's a real performance issue too.

Card 21.5

FRONT — QUESTION

What is box fill, and why does it matter for safety?

BACK — ANSWER & MEMORY TIP

Answer: Box fill is the calculated volume required inside an electrical box based on the conductors, devices and fittings it contains; exceeding the box's rated capacity risks overheating and makes safe termination difficult.

Memory tip: An overfilled box is a real fire and workmanship hazard, not just a code technicality.

Card 21.6

FRONT — QUESTION

How does a device (such as a receptacle or switch) count in a box fill calculation?

BACK — ANSWER & MEMORY TIP

Answer: A device typically counts as two conductor volume allowances, based on the largest conductor connected to it.

Memory tip: One device counts as two conductors' worth of volume.

Card 21.7

FRONT — QUESTION

What is conduit fill, and what is the general maximum percentage for three or more conductors?

BACK — ANSWER & MEMORY TIP

Answer: Conduit fill is the percentage of a raceway's cross-sectional area occupied by conductors; the general limit for three or more conductors is 40%.

Memory tip: Forty percent maximum fill keeps conductors from overheating and eases pulling.

Card 21.8

FRONT — QUESTION

What is the difference between a raceway and a cable, in general code terms?

BACK — ANSWER & MEMORY TIP

Answer: A raceway is an enclosed channel (such as conduit) into which conductors are pulled separately; a cable is a factory-assembled unit of conductors with its own outer covering.

Memory tip: Raceway holds conductors you pull in; cable already includes them.

Card 21.9

FRONT — QUESTION

Why does insulation type matter beyond just its temperature rating?

BACK — ANSWER & MEMORY TIP

Answer: Different insulation types have different resistance to moisture, oil, sunlight, or chemical exposure, and the wrong type for the environment can degrade prematurely even if the temperature rating is adequate.

Memory tip: Temperature rating alone doesn't guarantee suitability for the environment.

Card 21.10

FRONT — QUESTION

What must be checked when multiple current-carrying conductors share a single raceway over a long parallel run?

BACK — ANSWER & MEMORY TIP

Answer: Whether ampacity derating for conductor count applies, since heat from adjacent conductors accumulates and reduces each conductor's safe current-carrying capacity.

Memory tip: More current-carrying neighbours in the same raceway, more derating required.

Card 21.11

FRONT — QUESTION

Why is an enclosure's rating (such as its NEMA or IP rating) relevant to conductor and cable selection nearby?

BACK — ANSWER & MEMORY TIP

Answer: Because the enclosure's environmental rating must match the actual conditions (wet, dusty, corrosive), and cable or conductor entries must maintain that same level of protection.

Memory tip: The whole assembly is only as protected as its weakest entry point.

Chapter 22: Branch Circuitry, Luminaires and Lighting Controls

7 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 22.1

FRONT — QUESTION

What defines a branch circuit?

BACK — ANSWER & MEMORY TIP

Answer: The conductors between the final overcurrent device protecting the circuit and the connected outlets or utilization equipment.

Memory tip: Branch circuit is the last leg, from the breaker to the actual load.

Card 22.2

FRONT — QUESTION

What must be considered when loading a branch circuit with multiple receptacles or fixed loads?

BACK — ANSWER & MEMORY TIP

Answer: That the total connected and continuous load does not exceed the circuit's rated capacity, applying the appropriate continuous-load derating where relevant.

Memory tip: Sum the loads, apply derating where needed, compare to the circuit rating.

Card 22.3

FRONT — QUESTION

Why are dwelling receptacles typically required to be tamper-resistant?

BACK — ANSWER & MEMORY TIP

Answer: To reduce the risk of injury, particularly to young children, from inserting foreign objects into the receptacle's contacts.

Memory tip: Tamper-resistant receptacles are a targeted child-safety measure.

Card 22.4

FRONT — QUESTION

What is the general purpose of a three-way switch arrangement?

BACK — ANSWER & MEMORY TIP

Answer: To allow a single luminaire or lighting circuit to be controlled independently from two different locations.

Memory tip: Three-way switches let you turn a light on at one end and off at the other.

Card 22.5

FRONT — QUESTION

What is the function of an occupancy or motion sensor in lighting control?

BACK — ANSWER & MEMORY TIP

Answer: To automatically turn lighting on when occupancy is detected and off after a set period without detected motion, improving energy efficiency.

Memory tip: Occupancy sensors save energy by removing the human 'forgot to turn it off' factor.

Card 22.6

FRONT — QUESTION

Why must a dimmer be matched to the type of load it controls (such as LED versus incandescent)?

BACK — ANSWER & MEMORY TIP

Answer: Because different lighting technologies respond differently to dimming methods, and a mismatched dimmer can cause flickering, humming, or premature failure of the load.

Memory tip: Not every dimmer works with every lamp type; compatibility matters.

Card 22.7

FRONT — QUESTION

What is the significance of a luminaire's listed maximum lamp wattage or type?

BACK — ANSWER & MEMORY TIP

Answer: Exceeding the listed rating or using an incompatible lamp type can cause overheating and create a fire hazard, voiding the fixture's safety listing.

Memory tip: The fixture's rating is a hard limit, not a suggestion.

Chapter 23: HVAC Electrical Connections and Controls

5 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 23.1

FRONT — QUESTION

What is the electrician's typical role regarding HVAC equipment?

BACK — ANSWER & MEMORY TIP

Answer: To provide the power connection, disconnecting means, and control wiring to the unit, generally not the mechanical refrigerant or ductwork itself.

Memory tip: Electrician handles power and controls; mechanical trades handle refrigerant and ducting.

Card 23.2

FRONT — QUESTION

What do the nameplate abbreviations MCA and MOCP refer to on HVAC equipment?

BACK — ANSWER & MEMORY TIP

Answer: MCA is Minimum Circuit Ampacity, the minimum conductor ampacity required; MOCP is Maximum Overcurrent Protection, the largest protective device permitted for that circuit.

Memory tip: MCA sets the conductor floor; MOCP sets the protection ceiling.

Card 23.3

FRONT — QUESTION

Why must an HVAC unit's disconnecting means be located within sight of the equipment (or otherwise capable of being locked)?

BACK — ANSWER & MEMORY TIP

Answer: So a technician servicing the unit can be certain the power source is isolated and cannot be re-energized by someone else while work is underway.

Memory tip: Visible or lockable disconnect: no ambiguity about isolation during service.

Card 23.4

FRONT — QUESTION

Why does HVAC equipment often include both a compressor motor and fan motor controls in the same unit?

BACK — ANSWER & MEMORY TIP

Answer: Because both must be coordinated to prevent equipment damage (such as running the compressor without adequate airflow) and to allow correct sequencing and protection for each motor.

Memory tip: Multiple motors in one unit still each need proper coordination and protection.

Card 23.5

FRONT — QUESTION

Why is the MOCP value on an HVAC nameplate not simply calculated the same way as a typical single-motor branch circuit?

BACK — ANSWER & MEMORY TIP

Answer: Because HVAC equipment often contains multiple motors and other loads, and the manufacturer calculates a maximum protection value accounting for the combined equipment's startup and running characteristics.

Memory tip: The nameplate MOCP already accounts for the whole unit's real combined behaviour.

Chapter 24: Electric Heating Systems and Controls

5 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 24.1

FRONT — QUESTION

What are common types of electric heating encountered in residential and commercial work?

BACK — ANSWER & MEMORY TIP

Answer: Baseboard heaters, in-floor radiant heat, forced-air electric furnaces, and unit heaters, among others.

Memory tip: Electric heat comes in many forms, but all convert current directly to heat.

Card 24.2

FRONT — QUESTION

How is a heater's current draw typically calculated from its nameplate rating?

BACK — ANSWER & MEMORY TIP

Answer: Using P = E x I, current equals the heater's wattage divided by its rated voltage.

Memory tip: Heaters are resistive loads, so basic power formulas apply directly.

Card 24.3

FRONT — QUESTION

Why does sizing a heating circuit generally apply the continuous-load rule?

BACK — ANSWER & MEMORY TIP

Answer: Because heating loads commonly operate continuously for three hours or more, requiring the circuit to be sized at 125% of the load (equivalent to keeping the load at or below 80% of the device rating).

Memory tip: Heating loads are a textbook example of when the continuous-load rule applies.

Card 24.4

FRONT — QUESTION

What is the function of a thermostat in an electric heating circuit?

BACK — ANSWER & MEMORY TIP

Answer: To automatically control the heater's operation based on sensed temperature, cycling the load on and off (or modulating it) to maintain a setpoint.

Memory tip: Thermostat: temperature-based automatic control of the heating load.

Card 24.5

FRONT — QUESTION

What is a specific installation hazard associated with electric baseboard or in-floor heating?

BACK — ANSWER & MEMORY TIP

Answer: Combustible materials (such as furniture, drapery, or flooring materials) placed too close to or directly over heating elements can create a fire hazard from sustained heat exposure.

Memory tip: Electric heat's hazard is often about clearance to combustibles, not just electrical faults.

Chapter 25: Exit, Emergency and Life-Safety Lighting

4 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 25.1

FRONT — QUESTION

What is the general purpose of exit and emergency lighting systems?

BACK — ANSWER & MEMORY TIP

Answer: To ensure occupants can safely see and follow an escape route during a power outage or emergency, and to clearly mark exit paths at all times.

Memory tip: Exit lighting marks the way out; emergency lighting lets you see it even in the dark.

Card 25.2

FRONT — QUESTION

Why are exit and emergency lighting requirements typically governed by two different codes?

BACK — ANSWER & MEMORY TIP

Answer: The building code (or fire code) typically sets performance requirements (illumination levels, duration), while the electrical code governs the wiring and installation methods used to achieve them.

Memory tip: One code says what performance is needed; the other says how to wire it safely.

Card 25.3

FRONT — QUESTION

What must be true about the connection of self-contained emergency lighting (unit equipment)?

BACK — ANSWER & MEMORY TIP

Answer: It must be connected ahead of (unswitched by) any local lighting switch, so it activates automatically on a power failure regardless of switch position.

Memory tip: Emergency lighting can't be accidentally switched off; it connects ahead of the local switch.

Card 25.4

FRONT — QUESTION

What is typically required regarding the duration emergency lighting must remain operational after a power failure?

BACK — ANSWER & MEMORY TIP

Answer: A minimum duration (commonly a defined period such as 30 minutes) must be maintained to allow safe evacuation, as specified by the applicable building or fire code.

Memory tip: Emergency lighting needs a guaranteed minimum runtime, not just instant activation.

Chapter 26: Cathodic Protection Systems

3 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 26.1

FRONT — QUESTION

What does cathodic protection accomplish?

BACK — ANSWER & MEMORY TIP

Answer: It prevents or reduces corrosion of a metal structure (such as buried pipe or tanks) by making that structure the cathode of an electrochemical cell rather than the corroding anode.

Memory tip: Cathodic protection makes the structure the 'protected' side of a corrosion cell.

Card 26.2

FRONT — QUESTION

What are the two general types of cathodic protection systems?

BACK — ANSWER & MEMORY TIP

Answer: Sacrificial anode (galvanic) systems and impressed current systems.

Memory tip: Sacrificial anode: a metal 'sacrifices' itself. Impressed current: an external power source drives the protection.

Card 26.3

FRONT — QUESTION

What is the electrician's role, and what code governs impressed-current cathodic protection installations?

BACK — ANSWER & MEMORY TIP

Answer: The electrician typically installs and wires the rectifier and associated equipment for impressed-current systems, governed by the applicable section of the electrical code covering cathodic protection.

Memory tip: Impressed-current systems involve an electrical rectifier, which is squarely electrician's work.

Chapter 27: Motor Starters and Control Devices

11 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 27.1

FRONT — QUESTION

What is the difference between a motor's power circuit and its control circuit?

BACK — ANSWER & MEMORY TIP

Answer: The power circuit carries the actual current that runs the motor; the control circuit carries low-current signals that determine when the power circuit is switched on or off.

Memory tip: Power circuit does the work; control circuit makes the decisions.

Card 27.2

FRONT — QUESTION

How do you read a ladder diagram in terms of its basic left-to-right, top-to-bottom logic?

BACK — ANSWER & MEMORY TIP

Answer: Each horizontal 'rung' represents a control circuit path from one power rail to the other, with rungs read top to bottom in the order the logic is intended to be understood.

Memory tip: Think of ladder logic as a series of simple yes/no paths stacked vertically.

Card 27.3

FRONT — QUESTION

What is the difference between two-wire and three-wire motor control?

BACK — ANSWER & MEMORY TIP

Answer: Two-wire control (such as a simple thermostat or pressure switch) restarts the motor automatically once the controlling condition returns after a power loss; three-wire control (start-stop with a seal-in contact) requires the start button to be pressed again after a power loss.

Memory tip: Two-wire restarts on its own; three-wire waits for a human.

Card 27.4

FRONT — QUESTION

Why is three-wire control described as providing 'low-voltage protection'?

BACK — ANSWER & MEMORY TIP

Answer: Because the seal-in contact drops out during a power loss, so the motor does not automatically restart when power returns, protecting against unexpected startup.

Memory tip: Low-voltage protection means the motor stays off until manually restarted.

Card 27.5

FRONT — QUESTION

What are common motor starting methods used to reduce inrush current?

BACK — ANSWER & MEMORY TIP

Answer: Across-the-line (full-voltage) starting, reduced-voltage starting (such as autotransformer or wye-delta starting), and soft starters.

Memory tip: Big motors often need a gentler start than simply throwing the full-voltage switch.

Card 27.6

FRONT — QUESTION

What is the purpose of an overload relay in a motor starter?

BACK — ANSWER & MEMORY TIP

Answer: To protect the motor itself from sustained overcurrent (overload) conditions, distinct from the branch circuit's short-circuit protection.

Memory tip: Overload protects the motor; the branch device protects against faults.

Card 27.7

FRONT — QUESTION

How is a forward-reverse control circuit typically prevented from energizing both directions at once?

BACK — ANSWER & MEMORY TIP

Answer: Through mechanical and/or electrical interlocking, ensuring the forward and reverse contactors cannot both close simultaneously.

Memory tip: Interlocking prevents a direct short between opposing directions.

Card 27.8

FRONT — QUESTION

How is stop-device wiring generally arranged in a motor control circuit, and why?

BACK — ANSWER & MEMORY TIP

Answer: Stop devices are wired normally closed and in series, so that any single stop device opening will de-energize the entire control circuit.

Memory tip: Any one stop, in series, must be able to shut the whole thing down.

Card 27.9

FRONT — QUESTION

How is start-device wiring generally arranged, and why, compared to stop devices?

BACK — ANSWER & MEMORY TIP

Answer: Start devices are wired normally open and in parallel, so that any one of multiple start locations can initiate the circuit.

Memory tip: Any one start, in parallel, should be able to start the circuit.

Card 27.10

FRONT — QUESTION

What does 'sizing the motor circuit' generally require beyond simply reading the motor's nameplate current?

BACK — ANSWER & MEMORY TIP

Answer: Referring to the applicable code table for full-load current by horsepower and voltage, then applying the correct percentage factors for conductors, overload protection, and branch-circuit protection.

Memory tip: Installation sizing draws from code tables, not just nameplate numbers alone.

Card 27.11

FRONT — QUESTION

Why must overload protection be selected based on the motor's actual nameplate current rather than a table value?

BACK — ANSWER & MEMORY TIP

Answer: Because overload protection is meant to protect that specific motor from its own actual operating characteristics, which the nameplate reflects more precisely than a generic table.

Memory tip: Overloads are tuned to the specific motor; conductors and breakers are tuned to the table.

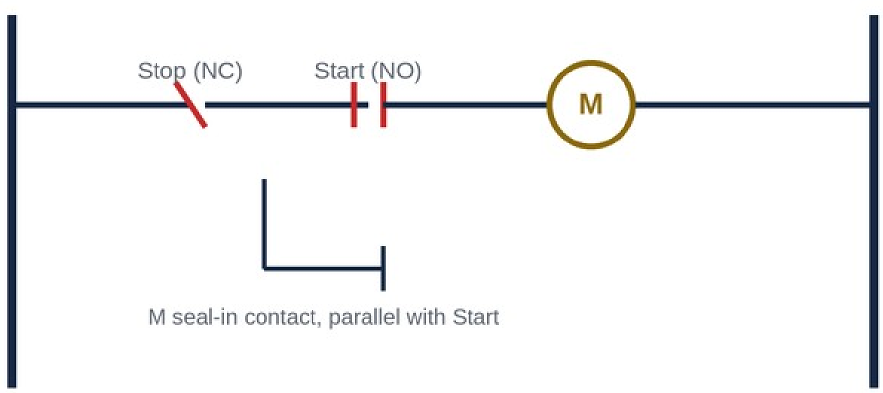

Figure FC-27.1 — A basic three-wire start-stop control rung with seal-in contact.

Chapter 28: AC and DC Drives

5 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 28.1

FRONT — QUESTION

What is the basic function of a variable-frequency drive (VFD)?

BACK — ANSWER & MEMORY TIP

Answer: It controls a motor's speed and torque by varying the frequency and voltage of the power supplied to it, rather than running the motor at a fixed line frequency.

Memory tip: VFD: vary the frequency, vary the motor speed.

Card 28.2

FRONT — QUESTION

Why do variable-frequency drives save energy compared to fixed-speed operation with mechanical throttling?

BACK — ANSWER & MEMORY TIP

Answer: Because they allow a motor to run only as fast as needed for the actual load (such as a fan or pump), rather than running at full speed and wasting energy through mechanical throttling or dampers.

Memory tip: Match the motor's speed to the actual need, and you save the wasted energy.

Card 28.3

FRONT — QUESTION

What installation problems can a VFD introduce that a standard fixed-speed motor circuit does not have?

BACK — ANSWER & MEMORY TIP

Answer: Electrical noise, harmonics on the supply, and potential bearing currents in the motor from the drive's switching behaviour.

Memory tip: VFDs bring real benefits but also introduce new electrical side effects to manage.

Card 28.4

FRONT — QUESTION

How does a soft starter differ from a variable-frequency drive?

BACK — ANSWER & MEMORY TIP

Answer: A soft starter reduces voltage during startup to limit inrush current but does not continuously vary frequency for ongoing speed control the way a VFD does.

Memory tip: Soft starter: gentle start only. VFD: continuous speed control throughout operation.

Card 28.5

FRONT — QUESTION

What is a basic function of a DC drive, in general terms?

BACK — ANSWER & MEMORY TIP

Answer: It controls the speed and torque of a DC motor by varying the voltage or current supplied to it, similar in concept to how a VFD controls an AC motor.

Memory tip: DC drive does for DC motors roughly what a VFD does for AC motors.

Chapter 29: Electric Motors

8 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 29.1

FRONT — QUESTION

How does a three-phase induction motor produce torque, in basic terms?

BACK — ANSWER & MEMORY TIP

Answer: The rotating magnetic field created by the three-phase stator windings induces current in the rotor, and the interaction between the two fields produces torque.

Memory tip: No direct electrical connection to the rotor is needed; induction does the work.

Card 29.2

FRONT — QUESTION

What is slip, in the context of an induction motor?

BACK — ANSWER & MEMORY TIP

Answer: The difference between synchronous speed and actual rotor speed, expressed as a percentage of synchronous speed, and it is what allows torque to be produced.

Memory tip: Zero slip means zero torque in a standard induction motor.

Card 29.3

FRONT — QUESTION

What is the general difference between single-phase and three-phase induction motors in terms of starting?

BACK — ANSWER & MEMORY TIP

Answer: Three-phase motors are inherently self-starting due to their naturally rotating magnetic field; single-phase motors require an additional starting mechanism (such as a start winding and capacitor) to begin rotation.

Memory tip: Three-phase starts itself; single-phase needs help getting going.

Card 29.4

FRONT — QUESTION

How does a DC motor's basic operating principle differ from an induction motor's?

BACK — ANSWER & MEMORY TIP

Answer: A DC motor typically uses a commutator and brushes (or electronic commutation) to maintain torque direction as the rotor turns, rather than relying purely on an externally rotating field.

Memory tip: DC motors need a mechanism to keep switching current direction inside the rotor as it spins.

Card 29.5

FRONT — QUESTION

What key information does a motor's nameplate provide for correct application and protection?

BACK — ANSWER & MEMORY TIP

Answer: Voltage, full-load current, horsepower, service factor, frame size, and other data needed to correctly size conductors, protection, and verify suitability for the application.

Memory tip: The nameplate is the primary source of truth for sizing that specific motor.

Card 29.6

FRONT — QUESTION

What is single-phasing, and why is it a hazard to a three-phase motor?

BACK — ANSWER & MEMORY TIP

Answer: Single-phasing occurs when one of the three supply phases to a motor is lost while running, causing the motor to draw excessive current on the remaining phases and overheat, potentially leading to failure.

Memory tip: Losing one phase doesn't stop the motor, it overheats it on the phases still connected.

Card 29.7

FRONT — QUESTION

What routine electrical test helps detect motor winding insulation degradation before failure?

BACK — ANSWER & MEMORY TIP

Answer: An insulation-resistance test using a megohmmeter, performed with the motor de-energized and isolated.

Memory tip: A megger test on motor windings can catch trouble before it becomes a failure.

Card 29.8

FRONT — QUESTION

Why must a motor be de-energized and isolated before performing an insulation-resistance test?

BACK — ANSWER & MEMORY TIP

Answer: Because the megohmmeter applies its own test voltage, and testing a motor that is still connected to a live source could produce inaccurate readings or create a hazard.

Memory tip: Isolate first; the megger needs a clean, de-energized subject to test properly.

Chapter 30: Automated and Programmable Control Systems

7 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 30.1

FRONT — QUESTION

Why did the programmable logic controller (PLC) largely replace traditional relay logic in industrial control?

BACK — ANSWER & MEMORY TIP

Answer: PLCs offer easier reprogramming, more compact hardware, better diagnostics, and greater flexibility compared to rewiring physical relay panels for every logic change.

Memory tip: Same control logic, but software-changeable instead of rewiring relays.

Card 30.2

FRONT — QUESTION

What are the basic parts of a PLC system?

BACK — ANSWER & MEMORY TIP

Answer: A central processing unit (CPU), input modules, output modules, and a power supply, working together to read inputs, execute logic, and drive outputs.

Memory tip: Inputs in, logic processed, outputs out, on a repeating cycle.

Card 30.3

FRONT — QUESTION

What is the scan cycle of a PLC?

BACK — ANSWER & MEMORY TIP

Answer: The repeating process where the PLC reads all inputs, executes the control program, then updates all outputs, continuously repeating at high speed.

Memory tip: Read, process, write, repeat, very fast, over and over.

Card 30.4

FRONT — QUESTION

What is ladder logic programming, in basic terms?

BACK — ANSWER & MEMORY TIP

Answer: A graphical programming method that visually resembles electrical relay ladder diagrams, making it intuitive for electricians transitioning from relay-based control.

Memory tip: Ladder logic mimics the relay diagrams electricians already know how to read.

Card 30.5

FRONT — QUESTION

What do HMIs and industrial networks add to a basic PLC control system?

BACK — ANSWER & MEMORY TIP

Answer: An HMI (human-machine interface) provides an operator interface to monitor and control the process; networks allow multiple PLCs and devices to communicate and coordinate as part of a larger system.

Memory tip: HMI: how a person interacts with it. Network: how machines talk to each other.

Card 30.6

FRONT — QUESTION

Why is a hard-wired safety circuit still required in many PLC-controlled systems, rather than relying on the PLC program alone for critical safety functions?

BACK — ANSWER & MEMORY TIP

Answer: Because a hard-wired safety circuit provides a fail-safe path independent of the PLC's software, ensuring critical stops function even if the program has a fault or the PLC itself fails.

Memory tip: Safety-critical stops shouldn't depend entirely on software that could fail or have a bug.

Card 30.7

FRONT — QUESTION

What is the first general step in troubleshooting a malfunctioning PLC-controlled system?

BACK — ANSWER & MEMORY TIP

Answer: Verifying inputs and outputs at the hardware level (are sensors reading correctly, are outputs actually energizing) before assuming the program logic itself is at fault.

Memory tip: Check the hardware inputs/outputs first, before blaming the software logic.

Chapter 31: Fire Alarm and Security Systems

7 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 31.1

FRONT — QUESTION

What is the general division of responsibility between codes governing fire alarm systems?

BACK — ANSWER & MEMORY TIP

Answer: The fire code and building code typically set performance and design requirements, while the electrical code governs the wiring methods and installation practices used.

Memory tip: One code says what the system must achieve; the other says how to wire it.

Card 31.2

FRONT — QUESTION

What is the basic difference between a conventional and an addressable fire alarm system?

BACK — ANSWER & MEMORY TIP

Answer: A conventional system identifies alarms by zone only; an addressable system identifies the exact individual device that triggered the alarm.

Memory tip: Conventional: 'somewhere in this zone.' Addressable: 'this exact device.'

Card 31.3

FRONT — QUESTION

What are the basic parts of a fire alarm system?

BACK — ANSWER & MEMORY TIP

Answer: Initiating devices (such as smoke detectors and pull stations), the fire alarm control panel, notification appliances (horns and strobes), and the interconnecting wiring.

Memory tip: Detect, process at the panel, then notify, all tied together by wiring.

Card 31.4

FRONT — QUESTION

What is an end-of-line resistor used for in fire alarm wiring, and what does it enable?

BACK — ANSWER & MEMORY TIP

Answer: It provides supervision of the circuit, allowing the panel to detect an open circuit (wiring fault) even when no alarm condition is present.

Memory tip: The end-of-line resistor lets the panel 'see' the wire is intact, even at rest.

Card 31.5

FRONT — QUESTION

What is the key difference between Class A and Class B fire alarm wiring in terms of fault tolerance?

BACK — ANSWER & MEMORY TIP

Answer: Class A wiring loops back to the panel, so a single break in the wiring does not disable devices beyond the break; Class B wiring does not loop back, so a single break can disable everything downstream of it.

Memory tip: Class A survives one break; Class B does not.

Card 31.6

FRONT — QUESTION

What general wiring practice separation is required between fire alarm circuits and other building wiring?

BACK — ANSWER & MEMORY TIP

Answer: Fire alarm circuits generally require separation from power and other low-voltage wiring to prevent interference and maintain circuit integrity and supervision.

Memory tip: Fire alarm wiring needs its own space, away from unrelated circuits.

Card 31.7

FRONT — QUESTION

What is the general function of an access control system, distinct from an intrusion (burglar) alarm?

BACK — ANSWER & MEMORY TIP

Answer: Access control manages and restricts who can enter specific areas (such as via card readers), while an intrusion alarm detects unauthorized entry or presence.

Memory tip: Access control lets the right people in; alarm systems detect the wrong people getting in.

[FIGURE FC-31.1 — ILLUSTRATION PLACEHOLDER]

Caption: A fire alarm control panel with labelled zones and an addressable device on a pull-station circuit.

Placement: Beside the addressable-vs-conventional card group.

Image generation prompt: *Flat technical illustration of a fire alarm control panel front face with labelled zone indicators and a small inset showing an addressable smoke detector with a unique device address, clean instructional diagram style, navy and red brand colours, no photorealism, textbook quality.*

Chapter 32: Voice, Data and Communication Systems

5 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 32.1

FRONT — QUESTION

What is structured cabling, in general terms?

BACK — ANSWER & MEMORY TIP

Answer: A standardized, organized cabling infrastructure design (such as for data, voice and video) that supports predictable performance and easier future changes, rather than ad-hoc point-to-point wiring.

Memory tip: Structured cabling is planned infrastructure, not a tangle of one-off runs.

Card 32.2

FRONT — QUESTION

What is the primary advantage of fibre-optic cable over copper cable for data transmission?

BACK — ANSWER & MEMORY TIP

Answer: Fibre offers much greater bandwidth and distance capability, and is immune to electromagnetic interference, unlike copper conductors.

Memory tip: Fibre: more data, farther, and immune to electrical noise.

Card 32.3

FRONT — QUESTION

What is Power over Ethernet (PoE), and what does it allow?

BACK — ANSWER & MEMORY TIP

Answer: A technology that delivers both data and low-voltage DC power over the same structured cabling, allowing devices like cameras or access points to be powered without a separate electrical circuit.

Memory tip: PoE: data and power together, on the same cable.

Card 32.4

FRONT — QUESTION

What is a nurse call system, in general purpose?

BACK — ANSWER & MEMORY TIP

Answer: A communication system, typically in healthcare facilities, allowing patients to summon staff assistance and allowing staff to communicate with a central station or each other.

Memory tip: Nurse call is a dedicated communication system for patient-to-staff assistance.

Card 32.5

FRONT — QUESTION

Why must bonding and separation practices for communication cabling be specifically addressed in the code, distinct from power wiring?

BACK — ANSWER & MEMORY TIP

Answer: Because low-voltage communication systems are sensitive to induced noise from power circuits, and improper bonding or insufficient separation can degrade signal quality or introduce safety hazards.

Memory tip: Communication cabling has its own separation and bonding needs, driven by signal sensitivity.

Chapter 33: Building Automation and Integrated Control

4 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 33.1

FRONT — QUESTION

What is a building automation system (BAS), in general terms?

BACK — ANSWER & MEMORY TIP

Answer: An integrated control system that monitors and manages a building's mechanical, electrical, lighting and other systems from a centralized platform.

Memory tip: BAS ties multiple building systems together under one coordinated control layer.

Card 33.2

FRONT — QUESTION

What are common communication protocols encountered in building automation systems?

BACK — ANSWER & MEMORY TIP

Answer: Protocols such as BACnet and Modbus are commonly used to allow different manufacturers' equipment to communicate within a BAS.

Memory tip: Protocols are the common 'language' that lets different brands of equipment talk to each other.

Card 33.3

FRONT — QUESTION

How does lighting control typically integrate into a broader building automation system?

BACK — ANSWER & MEMORY TIP

Answer: Lighting can be scheduled, dimmed, and coordinated with occupancy and daylight sensors as part of the overall automated building strategy, rather than operating as an isolated system.

Memory tip: In a BAS, lighting becomes one coordinated piece of a larger automated system.

Card 33.4

FRONT — QUESTION

What is the electrician's general role in a building automation system project?

BACK — ANSWER & MEMORY TIP

Answer: Installing the wiring, sensors, controllers and related devices per the system design, and often coordinating with system integrators or controls specialists for programming and commissioning.

Memory tip: Electrician installs the physical infrastructure; integration and programming are often a specialized, coordinated role.

Exam Toolkit: Cross-Reference and Terminology

19 cards in this chapter. Front: the question below. Back: the answer, plus a memory tip.

Card 34.1

FRONT — QUESTION

What Canadian standard is the base document for essentially all electrical installation work in Canada?

BACK — ANSWER & MEMORY TIP

Answer: The Canadian Electrical Code (CEC), Part I, CSA C22.1, though provinces adopt it with their own amendments.

Memory tip: CEC Part I is the national base; provincial amendments layer on top.

Card 34.2

FRONT — QUESTION

What does the metric prefix 'kilo' represent, and give an example unit.

BACK — ANSWER & MEMORY TIP

Answer: One thousand ($\times10^3$); for example, a kilovolt (kV) equals 1,000 volts.

Memory tip: Kilo = thousand, the most common prefix in electrical work.

Card 34.3

FRONT — QUESTION

What does the metric prefix 'milli' represent, and give an example relevant to electrical safety?

BACK — ANSWER & MEMORY TIP

Answer: One thousandth ($\times10^{-3}$); for example, GFCI trip thresholds are measured in milliamperes (mA).

Memory tip: Milli = one-thousandth, critical for understanding shock-current thresholds.

Card 34.4

FRONT — QUESTION

What Canadian term is generally used where U.S. documentation might say 'ground wire' or 'grounding conductor' for the connection between equipment and the system?

BACK — ANSWER & MEMORY TIP

Answer: Bonding conductor (for equipment bonding) is the Canadian term generally used, keeping the grounding/bonding distinction clear, which U.S. terminology often blurs.

Memory tip: Canadian code is precise about grounding versus bonding; U.S. terms often use 'ground' for both.

Card 34.5

FRONT — QUESTION

What Canadian term corresponds to the U.S. term 'branch circuit breaker panel'?

BACK — ANSWER & MEMORY TIP

Answer: Panelboard is the standard Canadian code term.

Memory tip: Panelboard, not 'breaker box,' is the correct code terminology.

Card 34.6

FRONT — QUESTION

What is the general Canadian code term for what is sometimes called a 'raceway' generically in casual conversation?

BACK — ANSWER & MEMORY TIP

Answer: Raceway remains the correct general term in the CEC as well, encompassing conduit, tubing, and similar enclosed wiring channels.

Memory tip: Raceway is both the casual and the correct code term in this case.

Card 34.7

FRONT — QUESTION

Roughly which CEC section governs conductor ampacity tables?

BACK — ANSWER & MEMORY TIP

Answer: Section 4 of the CEC covers conductors, including ampacity tables and related sizing rules.

Memory tip: Section 4: conductors and their ampacity.

Card 34.8

FRONT — QUESTION

Roughly which CEC section governs overcurrent protection?

BACK — ANSWER & MEMORY TIP

Answer: Section 14 of the CEC covers protection and control, including overcurrent devices.

Memory tip: Section 14: protection and control.

Card 34.9

FRONT — QUESTION

Roughly which CEC section governs grounding and bonding?

BACK — ANSWER & MEMORY TIP

Answer: Section 10 of the CEC covers grounding and bonding requirements.

Memory tip: Section 10: grounding and bonding, the section most exam questions on this topic reference.

Card 34.10

FRONT — QUESTION

Roughly which CEC section governs services and service equipment?

BACK — ANSWER & MEMORY TIP

Answer: Section 6 of the CEC covers services and service equipment.

Memory tip: Section 6: services, the starting point of the whole system.

Card 34.11

FRONT — QUESTION

Roughly which CEC section covers motors and generators?

BACK — ANSWER & MEMORY TIP

Answer: Section 28 of the CEC covers motors and generators.

Memory tip: Section 28: everything about motor circuit sizing and protection.

Card 34.12

FRONT — QUESTION

Roughly which CEC section covers hazardous locations?

BACK — ANSWER & MEMORY TIP

Answer: Section 18 of the CEC covers hazardous locations.

Memory tip: Section 18: classified (hazardous) locations.

Card 34.13

FRONT — QUESTION

What is the general exam-day strategy for handling a question containing unfamiliar or unusually specific terminology?

BACK — ANSWER & MEMORY TIP

Answer: Focus on the underlying principle being tested rather than the specific term; most unfamiliar terms are variations of a concept you already know from a different angle.

Memory tip: Strip away unfamiliar wording and look for the concept underneath.

Card 34.14

FRONT — QUESTION

Why is time management considered as important as technical knowledge on a long, multi-part trade exam?

BACK — ANSWER & MEMORY TIP

Answer: Because running out of time before reaching every question guarantees lost marks regardless of how well-prepared a candidate is on the material not reached.

Memory tip: Perfect knowledge with poor pacing still loses marks.

Card 34.15

FRONT — QUESTION

What is a reliable general strategy when two answer options both seem technically plausible?

BACK — ANSWER & MEMORY TIP

Answer: Re-read the question stem for a qualifier (such as 'minimum,' 'first,' or 'most appropriate') that distinguishes which of the two plausible answers is actually correct.

Memory tip: When two options both look right, the stem's exact wording is the tiebreaker.

Card 34.16

FRONT — QUESTION

What does the metric prefix 'mega' represent, relevant to insulation testing?

BACK — ANSWER & MEMORY TIP

Answer: One million ($\times 10^6$); insulation resistance is commonly measured in megohms (M-ohm) using a megohmmeter.

Memory tip: Mega = million, the scale insulation resistance is normally measured in.

Card 34.17

FRONT — QUESTION

Why is it useful to know which CEC section a topic falls under, even if exact section numbers aren't required on every question?

BACK — ANSWER & MEMORY TIP

Answer: Because it helps organize study time around the code's own structure and speeds up real-world code navigation, a skill the exam indirectly rewards.

Memory tip: Knowing the section map makes both studying and real code lookups faster.

Card 34.18

FRONT — QUESTION

What is a practical final-review technique using this card deck in the last week before the exam?

BACK — ANSWER & MEMORY TIP

Answer: Shuffle the deck and go through it in random order, since chapter-ordered review can create false confidence from sequence memory rather than true recall.

Memory tip: Shuffled review tests real recall, not just 'what comes next' pattern memory.

Card 34.19

FRONT — QUESTION

Why do Canadian trade exams sometimes include a mix of metric and imperial-derived values (such as circular mils)?

BACK — ANSWER & MEMORY TIP

Answer: Because some legacy imperial-based methods (like circular mils and K-factor voltage drop) remain in common industry use alongside the code's primarily metric framework, and candidates are expected to be fluent in both.

Memory tip: Real jobsites mix metric and legacy imperial methods, so the exam does too.